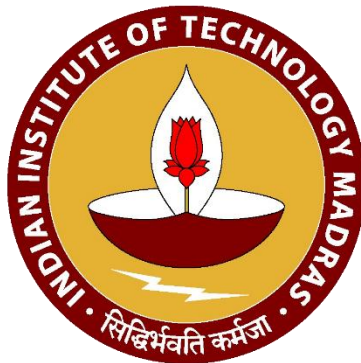


DA5400: Foundations of Machine Learning

Topic 9: Support Vector Machines

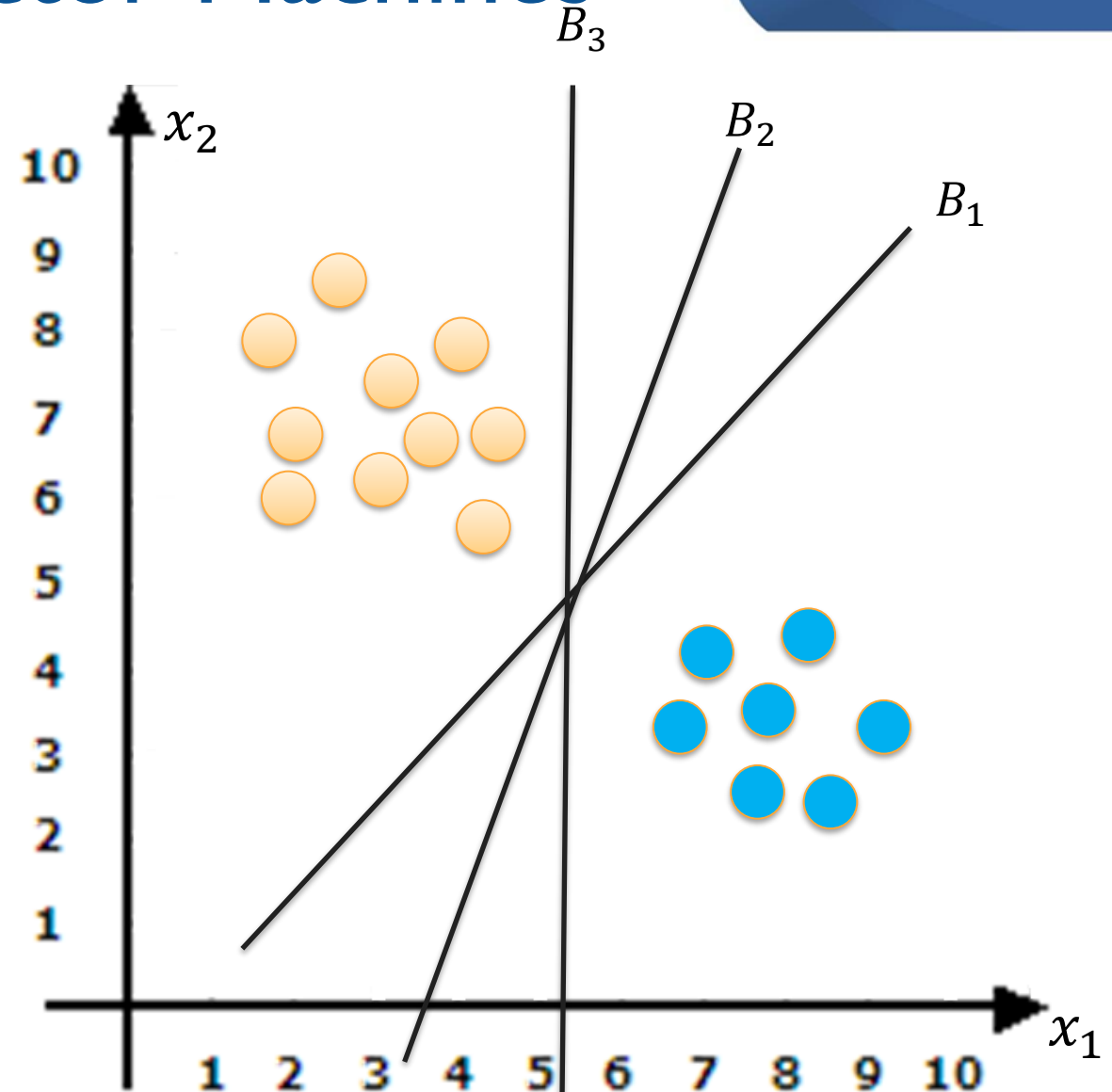


Contents

- Motivation for SVM
- Support Vectors & Margin
- Objective of SVM
- Hard Vs Soft Margin SVM

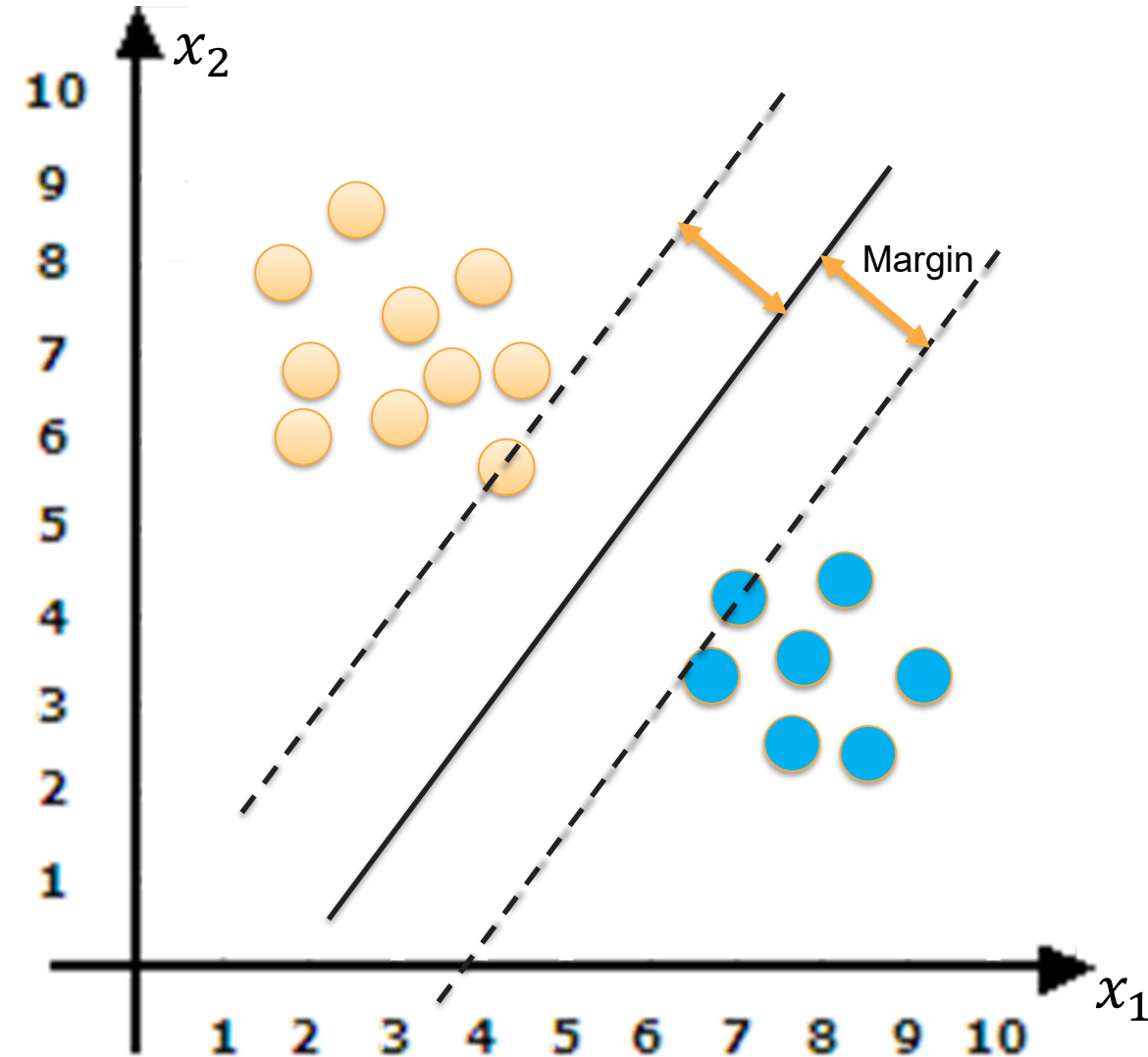
Motivation for Support Vector Machines

- Suppose two classes of data is linearly separable
- **Issue:** Many separating hyperplanes may exist
- **Q:** Which hyperplanes to choose among them? On what basis?
- **A:** Choose the hyperplane that generalises best to unseen data
- **SVM Idea:** Choose the hyperplane that leaves the largest separation between classes
- **Why:** A larger separation often gives better robustness on unseen data



Motivation for Support Vector Machines

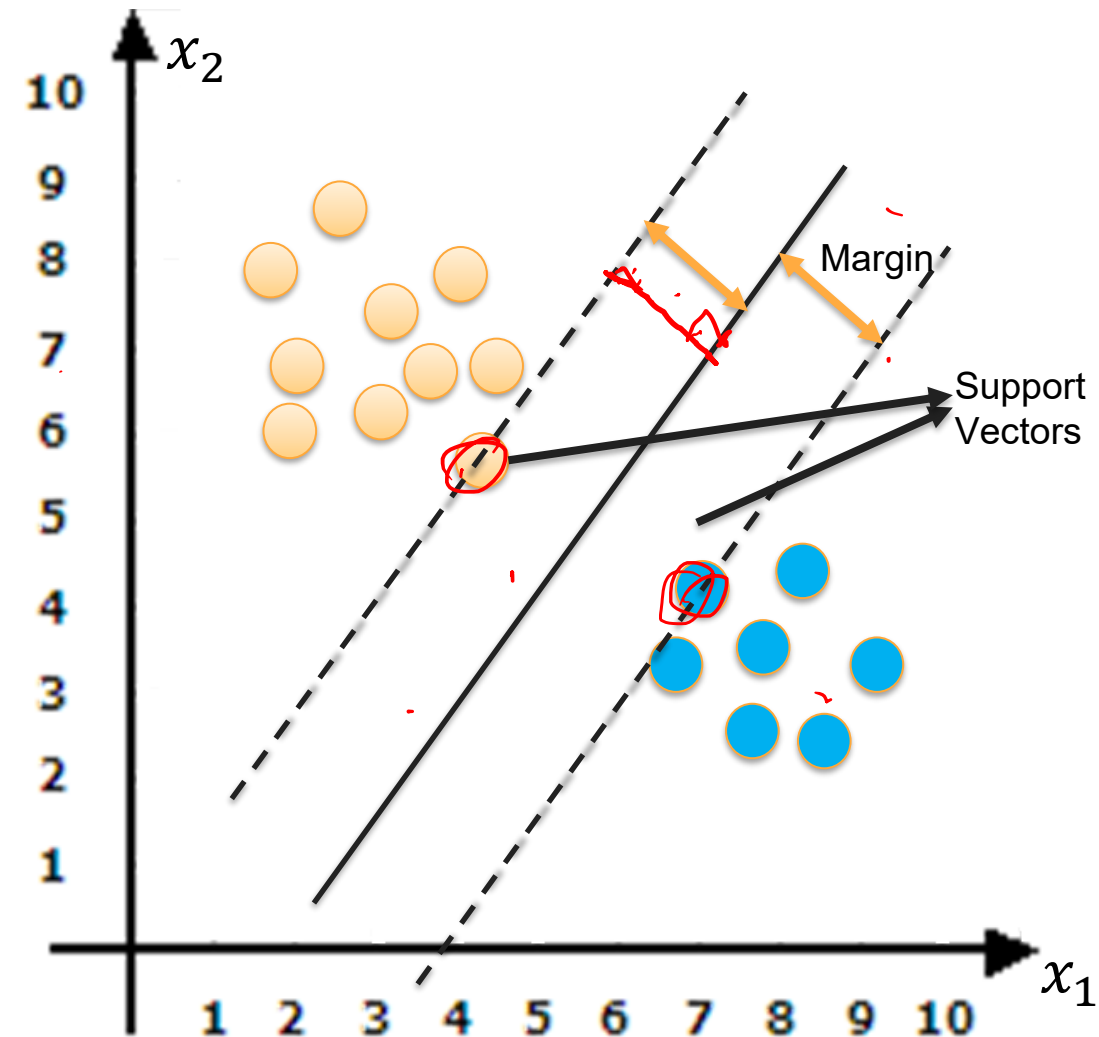
- Suppose two classes of data is linearly separable
- **Issue:** Many separating hyperplanes may exist
- **Q:** Which hyperplanes to choose among them? On what basis?
- **A:** Choose the hyperplane that generalises best to unseen data
- **SVM Idea:** Choose the hyperplane that leaves maximum margin on either side
- **Why:** A larger margin often gives better robustness on unseen data



Terminology: Support Vectors & Margin

- **Support Vectors:** Datapoints closest to the decision boundary
- **Margin:** Perpendicular distance between the support vectors of a class
- **Equation of decision boundary:**
 $w^T x + b = 0$
- Perpendicular distance of a point x_i from the boundary: $\frac{|w^T x_i + b|}{\|w\|}$
- We can show that support vectors satisfy:

$$w^T x + b = 1 \quad \text{or} \quad w^T x + b = -1$$



SVM: Objective

- Distance between support vectors and boundary:

$$\text{Margin} = \frac{1}{\|w\|}$$

- Total margin width on both sides: $\frac{2}{\|w\|}$

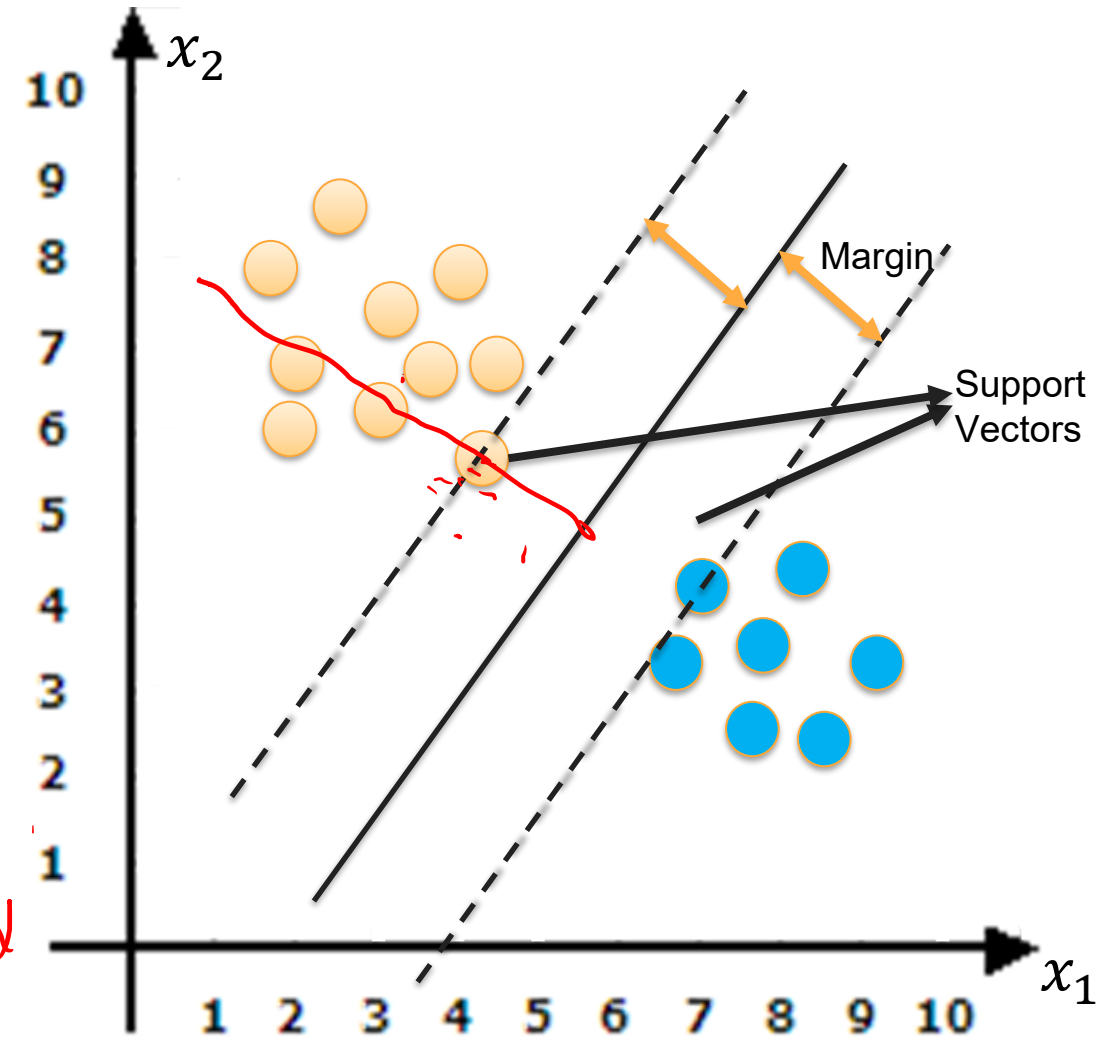
- Objective of SVM: Maximise total margin

$$\text{Max}_{w,b} \frac{2}{\|w\|}$$

$$(\text{or}) \quad \text{Min}_{w,b} \frac{1}{2} \|w\|$$

$$\text{s.t. } y_i(w^T x_i + b) \geq 1$$

$$y_i \in \{-1, +1\} \quad \forall i = 1, 2, \dots, N$$

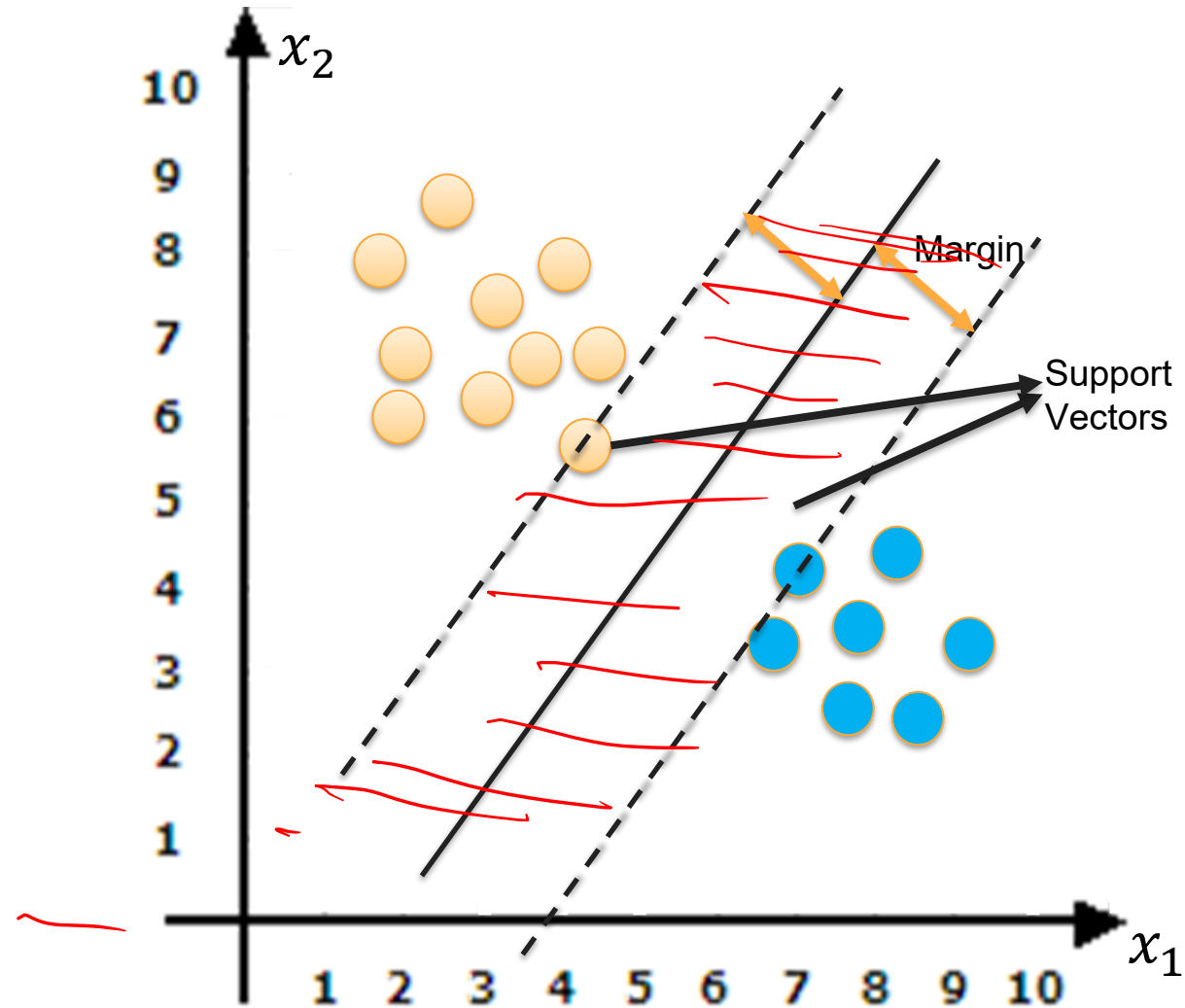


SVM: Intuition and Predictions

- **Intuition:** SVM learns mainly from boundary points, not all points equally
- **Prediction rule:**

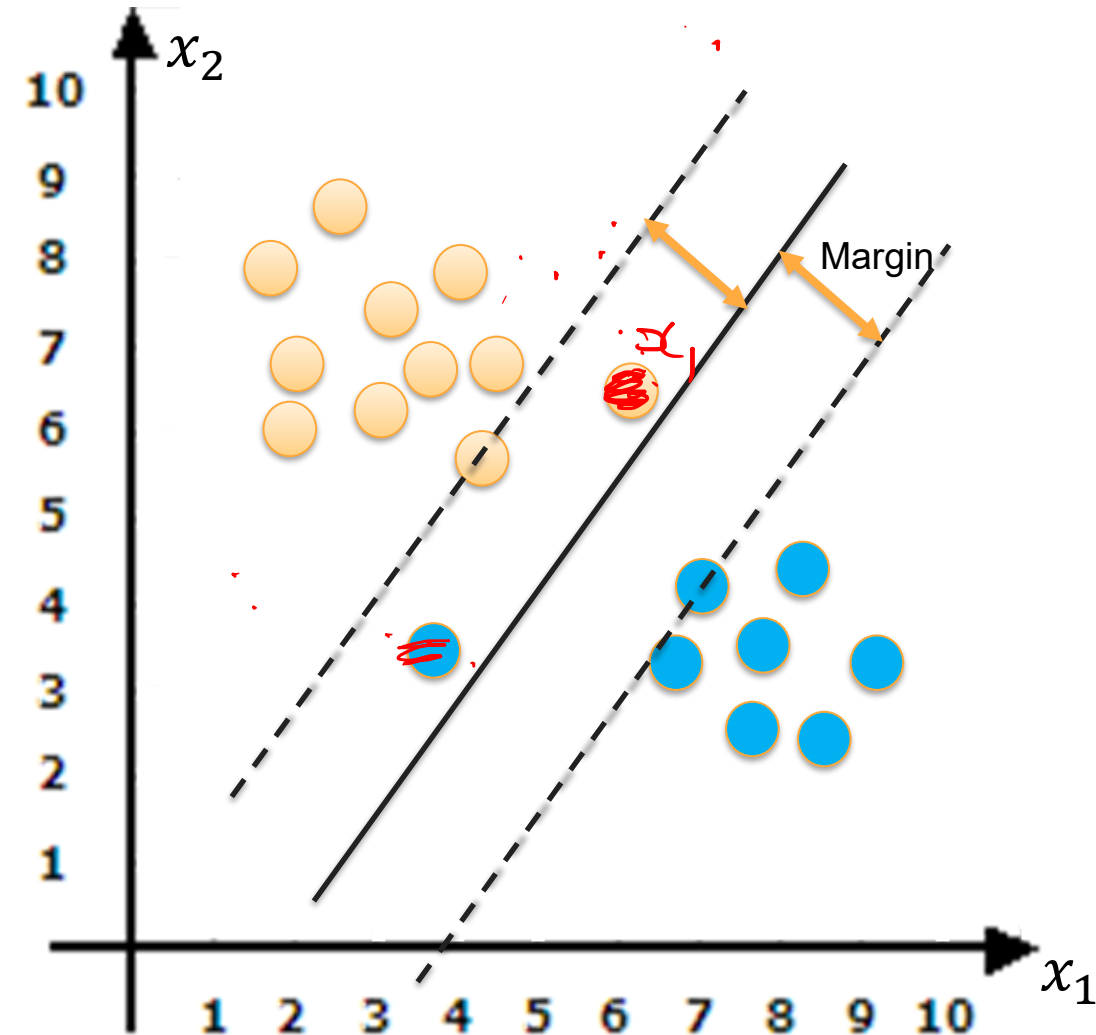
$$\hat{y} = \text{sign}(w^T x + b)$$

- This variant of SVM is referred to as Hard margin SVM
- **Issue:** In practice, classes might not be separable due to noise, overlap, outlier



Soft Margin SVM

- Soft margin SVM allows for controlled violations when classes are not fully separable
- **Intuition:** Finds a general decision boundary even if a points are inside the margin or even on the wrong side
- **Idea:** Introduce slack variables to measure how much a datapoint violates the ideal margin rule
- **Objective:** Soft Margin SVM balances between keeping the margin large and avoiding too many violations



Soft Margin SVM: Math Version

- Slack Variables: $\xi_i \forall i=1, 2, \dots, N$

- Values of slack variables and interpretation:

$\xi_i = 0$: No slack - No classification error / No margin violation

$0 < \xi_i < 1$: Correctly classified / Violates margin

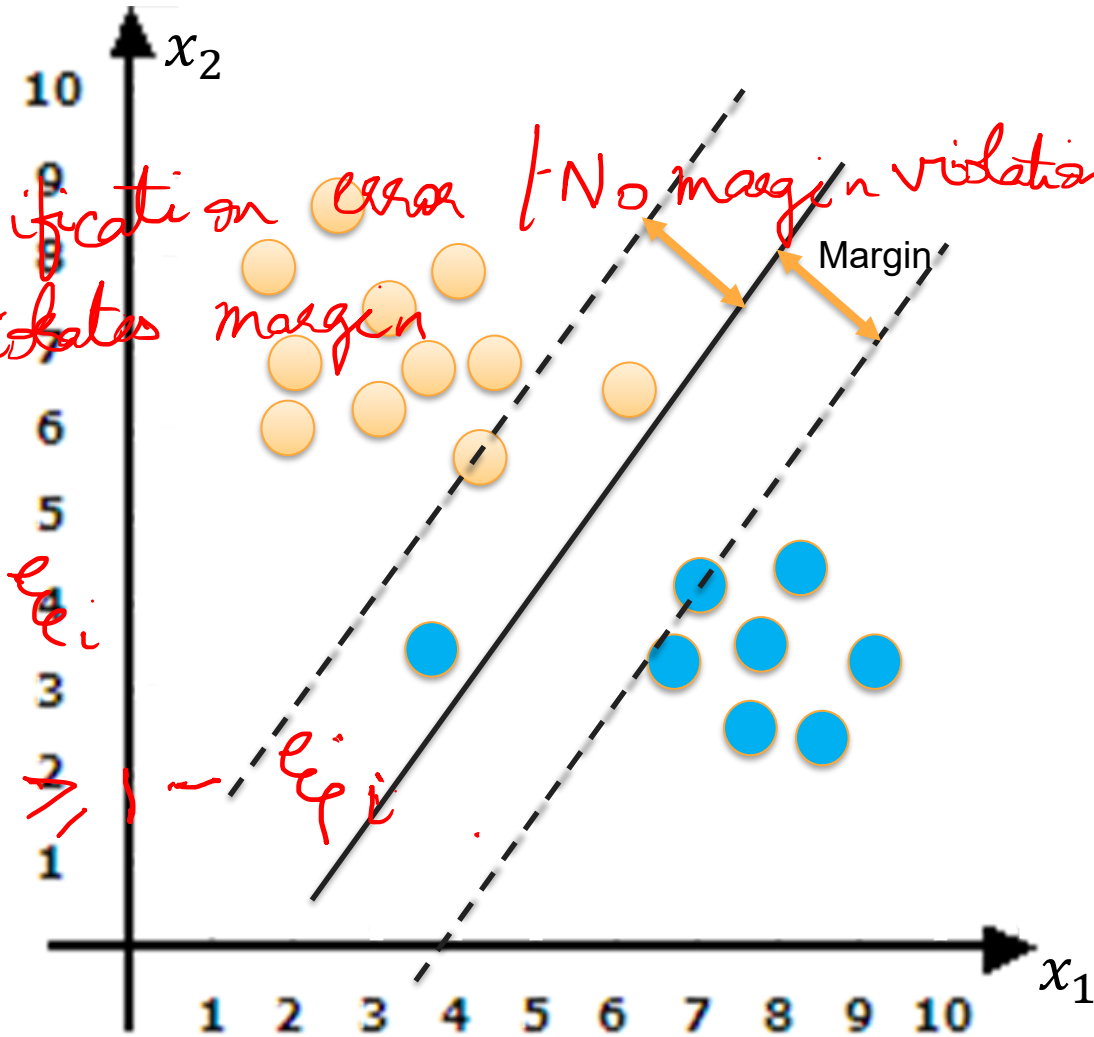
$\xi_i > 1$: Point is misclassified

- Objective:

$$\text{Min}_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i$$

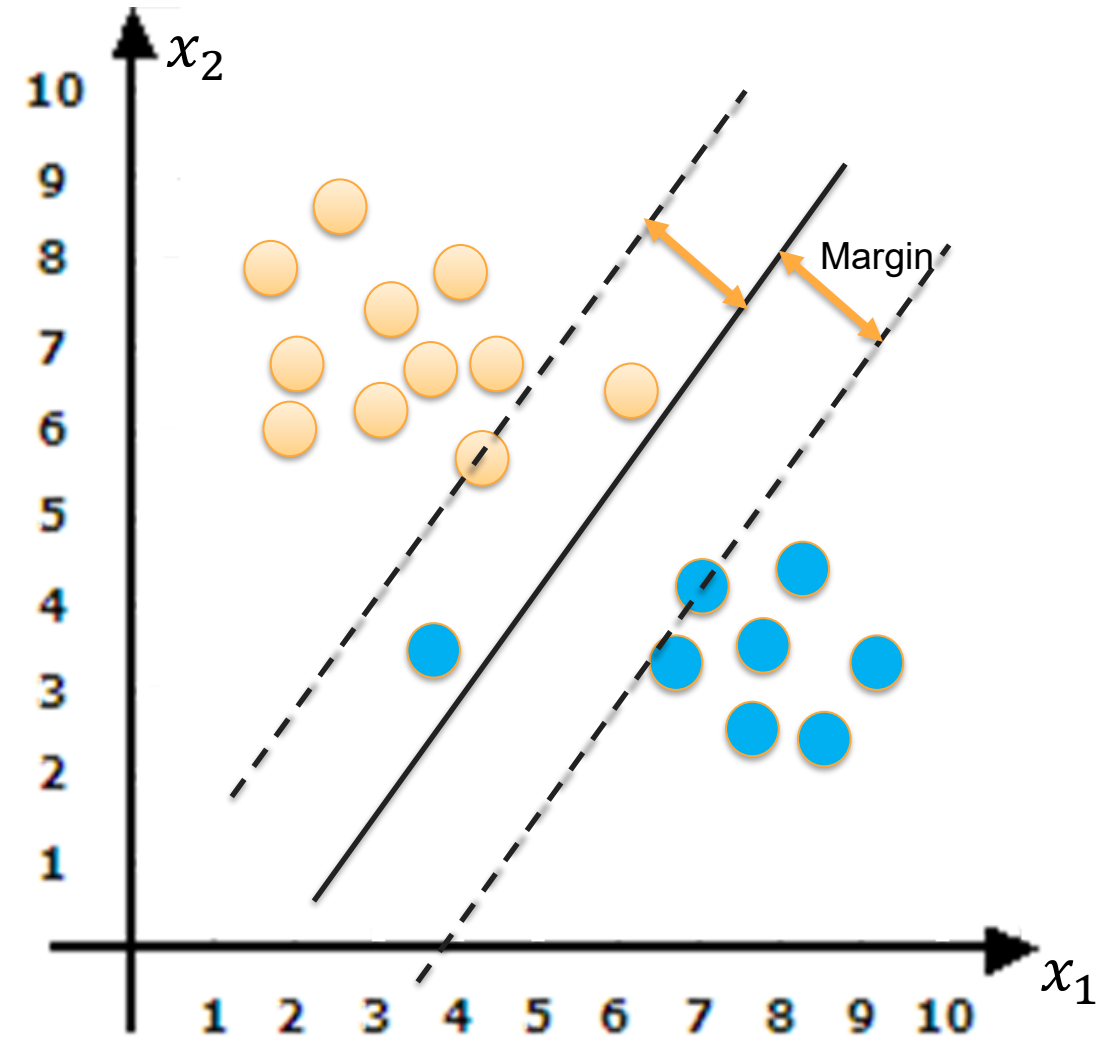
$$\text{s.t. } y_i(w^T x_i + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$



Soft Margin SVM: Value of C

- C allows for trade off between:
 - Larger margin (Generalisation)
 - Fewer classification errors (training error)
- Large C implies:
 - Stricter fit to training data
 - Weaker effective regularisation & generalisation
- Small C implies:
 - Tolerant fit and allows for more error on training data
 - Stronger regularisation & generalisation



SVM Practical Use & Extensions

- SVM works well in practice due to:
 - strong geometric foundation
 - robust for moderate-sized datasets
 - effective in high dimensions
- Limitations of SVMs:
 - Sensitive to feature scaling
 - Do not scale very well to very large datasets
 - Less interpretable compared to other classifiers
 - No probabilistic output is available
 - No natural extension to multi-class classification
- **Extensions:** Kernel SVM allows for extension to non-linear decision boundaries

```
operation == "MIRROR_X":  
    mirror_mod.use_x = True  
    mirror_mod.use_y = False  
    mirror_mod.use_z = False  
operation == "MIRROR_Y":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = True  
    mirror_mod.use_z = False  
operation == "MIRROR_Z":  
    mirror_mod.use_x = False  
    mirror_mod.use_y = False  
    mirror_mod.use_z = True
```

```
#selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
= ("Selected" + str(modifier_ob.name))  
mirror_ob.select = 0  
= bpy.context.selected_objects  
data.objects[one.name].select  
print("please select exactly one mirror")
```

WILLIAM C. LEE

```
def mirror(modifier):  
    #add mirror to the selected  
    #object -mirror_x, mirror_y,  
    #mirror_z  
    mirror_ob = bpy.context.selected_objects[0]  
    mirror_mod = modifier
```

THANK YOU